Digital preservation strategies for historical records in the age of AI and the metaverse

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Introduction

Digital preservation is a critical process aimed at ensuring the long-term accessibility and usability of digital records, particularly in the context of rapidly evolving technologies such as artificial intelligence (AI) and the metaverse. Effective strategies for digital preservation encompass organizational policies, technical frameworks and proactive management throughout the lifecycle of digital objects (Dečman and Vintar, 2013; Barrueco and Graells, 2021). The integration of AI can enhance these strategies by enabling automated preservation processes and improving metadata management, which is essential for maintaining the authenticity and integrity of digital records (Masenya, 2021). Moreover, the metaverse presents unique challenges and opportunities for digital preservation, as it blurs the lines between physical and digital assets, necessitating innovative approaches to ensure that historical records remain accessible and relevant in these new environments (Chu *et al.*, 2022). As organizations adapt to these changes, it is imperative to foster a culture of digital preservation that incorporates both technological advancements and the evolving needs of users (Aziera Alui *et al.*, 2022; Pasqui, 2024).

Historical records and digital challenges

The preservation of historical records faces significant challenges, primarily due to digital obsolescence and technological decay. Digital obsolescence refers to the rapid evolution of technology, which can render digital formats and storage media obsolete, thereby jeopardizing access to historical records (Sambo *et al.*, 2017; Muthoka *et al.*, 2021). For instance, in our library, we have faced significant challenges due to digital obsolescence, especially with older storage systems becoming incompatible with modern platforms. Our experience with upgrading these systems highlights the importance of regular audits and updates to preservation strategies, ensuring long-term access to our collections (Sambo *et al.*, 2017; Muthoka *et al.*, 2021). Moreover, technological decay, which encompasses the deterioration of storage media and the loss of data integrity over time, further complicates preservation efforts (Muthoka *et al.*, 2021). For instance, we recently undertook a project to update outdated storage media, only to realize that some of our digital records were no longer accessible. This issue prompted us to adopt AI tools that help predict which formats are at risk of becoming obsolete and automate the process of updating them. However, we still face challenges with funding and standardization, as preservation costs continue to rise. As noted, the transient nature of Web resources necessitates robust strategies to ensure long-term availability and access (Muthoka *et al.*, 2021). The integration of emerging technologies, such as blockchain, may offer potential solutions for enhancing data security and integrity in digital preservation, although these technologies also introduce new complexities (Huynh-The *et al.*, 2023). Thus, addressing these challenges requires a multifaceted approach that combines technological innovation with effective policy frameworks and training for preservation professionals (Sambo *et al.*, 2017; Muthoka *et al.*, 2021).

Artificial intelligence and its role in digital preservation

AI plays a transformative role in digital preservation, particularly through AI-driven tools for digitization and predictive models for record management. At the National Open University of Nigeria's Library, we have been testing AI tools like image recognition software for automating the digitization of historical records. From our experience, this has significantly reduced the time spent on manual processes, though we encountered initial challenges with training the AI on our specific archives. We have begun incorporating AI to assist with the digitization of our collections, particularly for large volumes of audiovisual materials. Through tools like machine learning algorithms, we have been able to significantly reduce the time it takes to process these materials. One example is our recent project digitizing old newspaper records. AI software helped with text recognition and categorization, allowing us to quickly create searchable databases for patrons. This experience has been largely positive, though we encountered initial difficulties with the AI's accuracy, which required manual corrections during the early phases of implementation.

In the metaverse, historical archives can be transformed into interactive experiences (Schellnack-Kelly, 2024). For instance, AI algorithms can analyze and categorize vast amounts of audiovisual content, ensuring that critical historical data is not lost and remains accessible for future generations (Schellnack-Kelly, 2024). Moreover, predictive models using AI can optimize record management by analyzing patterns in historical data to forecast future preservation needs. These models can identify at-risk records based on usage trends and environmental conditions, allowing for proactive preservation strategies. By leveraging machine learning techniques, organizations can enhance their ability to maintain the integrity and accessibility of historical records, addressing challenges posed by digital obsolescence and technological decay. Thus, the integration of AI in digital preservation not only streamlines processes but also strengthens the long-term viability of historical records.

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Preservation in the metaverse

The metaverse, a collective virtual shared space created by the convergence of virtually enhanced physical reality and physically persistent virtual reality, has significant implications for the preservation and accessibility of historical archives. It serves as a digital environment where users can interact with 3D representations of historical artifacts, documents and experiences, thereby redefining how historical records are displayed and engaged with (Barazzetti and Roncoroni, 2021). The relevance of the metaverse to historical archives lies in its potential to create immersive experiences that enhance public engagement and education regarding historical events and cultural heritage (Barazzetti and Roncoroni, 2021).

In the metaverse, historical archives can be transformed into interactive experiences, allowing users to explore and engage with historical records in ways that traditional formats cannot offer. For instance, we have observed how immersive environments can attract new audiences, although we are still refining how best to curate these experiences while maintaining historical accuracy (Barazzetti and Roncoroni, 2021). This immersive approach not only preserves the essence of historical records but also makes them more accessible to a broader audience, including those who may not have the opportunity to visit physical archives (Barazzetti and Roncoroni, 2021). Moreover, the metaverse allows for collaborative experiences where users can interact with historical records alongside others, facilitating discussions and shared learning experiences (Barazzetti and Roncoroni, 2021). This social aspect of the metaverse can enhance the educational value of historical archives, making them more engaging and relevant to contemporary audiences. By leveraging AI-driven tools, historical records can be digitized and indexed efficiently, ensuring that they are easily searchable and retrievable within these virtual environments (Colavizza *et al.*, 2019).

The integration of AI in the metaverse also supports the development of predictive models that can analyze user interactions with historical records, providing insights into how these records are used and perceived (Tasar, 2023). Such models can inform curatorial decisions, helping archivists and historians understand which aspects of history resonate most with users and how to present these narratives effectively in virtual spaces (Tasar, 2023). Furthermore, the metaverse presents opportunities for innovative storytelling techniques that can breathe new life into historical narratives. By combining historical data with immersive technologies, creators can craft compelling narratives that engage users emotionally and intellectually, thereby fostering a greater appreciation for history (Barazzetti and Roncoroni, 2021).

This approach not only preserves historical records but also revitalizes them, ensuring their relevance in an increasingly digital world. In conclusion, the metaverse offers a unique platform for the preservation and presentation of historical records, transforming how these records are accessed and experienced. By harnessing the power of immersive technologies and AI, historical archives can be preserved in ways that enhance their accessibility, engagement and educational value, ensuring that they continue to inform and inspire future generations.

Challenges and ethical considerations

As our library embraces AI and the metaverse, we are also navigating ethical concerns. One particular challenge we faced was data ownership when digitizing culturally sensitive historical records. Some of our collections involve records from indigenous communities, and we had to work closely with these communities to ensure that their rights and perspectives were respected. Our AI tools had to be customized to ensure that sensitive data was not inadvertently exposed or misused during the digitization process.

In addition, privacy concerns arose when using AI to analyze user data for improving our archives' engagement strategies. We took care to comply with regulations like the General Data Protection Regulation (GDPR) and implemented stricter privacy measures, ensuring that user interactions with our digital collections remain secure (Afwan Hariri, 2021).

Privacy concerns are paramount, particularly with the implementation of AI technologies that may inadvertently expose sensitive information. The GDPR has heightened awareness of the need for robust privacy measures, compelling institutions to rethink how they manage personal data within digital archives (Burke and Zavalina, 2020).

Furthermore, the digitization process itself can pose risks to the integrity of historical records, as errors during digitization may compromise authenticity and reliability (Balogun and Adjei, 2018). Ethically, institutions must balance the benefits of increased access to historical records with the potential for misuse or misrepresentation of data. This includes ensuring that digitized materials are presented in a context that maintains their historical significance and integrity (Li, 2024). As such, developing clear policies around data ownership, privacy and ethical considerations in digital preservation is essential for safeguarding both the records and the rights of individuals involved.

Conclusion

The future of digital preservation is poised to be significantly influenced by the integration of AI and metaverse technologies. As institutions increasingly adopt these technologies, several promising directions emerge. At the National Open University of Nigeria's Library, Our experiences show that while AI and the metaverse offer significant opportunities for advancing digital preservation, they also present unique challenges. We have seen first-hand how AI can enhance our digitization processes, but we have also faced difficulties with initial accuracy and resource allocation. The metaverse offers an exciting new way to engage with our archives, though it requires careful management to balance historical integrity with user immersion. Going forward, we plan to continue refining these technologies while also addressing ethical concerns and ensuring that our collections remain accessible for future generations (Huynh-The *et al.*, 2022; Rincón Soto and Sánchez León, 2022). In the metaverse, the potential for immersive experiences offers a transformative approach to how historical records are displayed and interacted with.

By creating virtual environments, users can engage with historical artifacts in ways that foster deeper understanding and appreciation (Amzat and Adewojo, 2023). This not only preserves the records but also revitalizes them, making them more relevant to contemporary audiences

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(Amzat and Adewojo, 2023). Additionally, the metaverse can facilitate collaborative learning experiences, allowing users to explore and discuss historical contexts collectively (Amzat and Adewojo, 2023). Moreover, the integration of blockchain technology within the metaverse can enhance data security and ownership transparency, addressing ethical concerns related to data privacy and historical integrity (Aloqaily *et al*., 2023; Yang *et al*., 2022). As these technologies evolve, ongoing research and development will be essential to navigate the challenges and opportunities they present, ensuring that digital preservation efforts remain effective and ethically sound in the rapidly changing digital landscape (Huynh-The *et al*., 2022; Yang *et al*., 2022).

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